

Niloy Saha

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EDUCATION

Hofstra University

Bachelor of Science, *Computer Science*

GPA: 3.5

Related Coursework: *Computing in Python, Advanced Data Structures, Algorithms and OOP, Software Engineering, Discrete Structures & Mathematics, Computer Operating Systems, Foundations of NLP & Text Mining*

New York, USA

Expected Graduation, May 2027

The Complete Full-Stack Web Development Bootcamp

Date Issued: 20 Aug 2025

By Udemy

Completion Certificate: <https://www.udemy.com/certificate/UC-60a6fb8e-10a6-4e5c-8f3b-b6ede6670c68/>

PROJECTS

Credit Card Fraudulent Detector | [GitHub Link](#)

- Built a credit card fraud detection system using **Python, Pandas, NumPy, and scikit-learn** on **50,000+** transactions with **8** key features, preprocessing and normalizing data, and handling class imbalance with **SMOTE**.
- Trained **Random Forest** and **Gradient Boosting classifiers**, achieving **99%+** ROC-AUC and **0.99** F1-score, and implemented a real-time **prediction function** for new transactions.
- Generated visual insights, including transaction amount and class distributions using **matplotlib**, to support model interpretability and feature importance analysis.

CourseCrafter AI | [GitHub Link](#) | [Site Link](#)

- Developed a full-stack **Node.js & Express** web app that generates structured AI learning courses from user-provided topics.
- Integrated **OpenAI API** and **YouTube Data API** with **Axios** to create dynamic course outlines (modules, lessons, and resources), optimizing backend performance and reducing API response times by **30%**.
- Delivered a modern, responsive UI using **EJS templating** and **Bootstrap**, ensuring seamless, accessible learning experiences across devices.

EXPERIENCE

Resident Success Representative, Hofstra University

January 2025 – April 2025

- Assisted and guided university residents, resolving housing and campus-related issues to ensure a safe and supportive living environment.
- Coordinated events, workshops, and community programs, fostering engagement and enhancing resident satisfaction across the dormitories.
- Collaborated with university staff to address student concerns, track incidents, and implement solutions, contributing to a well-organized and responsive residential community.

SKILLS

- Programming Languages:** Java, Python, C++, HTML/CSS, JavaScript, Dart
- Technical/Programmatic:** MongoDB, SQL, SQLite, SQLAlchemy, PostgreSQL, NoSQL, JQuery, React.js, Node.js, Express.js, Flask, EJS, Jinja, Git, Axios, Matplotlib, Pandas, Numpy, Scikit-Learn
- Concepts:** Data Structures and Algorithms, Object-Oriented Programming, Agile Methodologies, RESTful APIs, Natural Language Processing, Version Control, SDLC, Machine Learning
- Tools & Platforms:** GitHub, Visual Studio Code, PyCharm, Xcode, Postman, Unreal Engine, Jupyter Notebook

EXTRACURRICULAR ACTIVITIES

Sugar Realms — 3D Base-Building Game

February 2025 – May 2025

Hofstra University

Developer — Core game functionality and UI

- Built core base-building systems (resource collection, structure placement, upgrades) for Sugar Realms in **Unreal Engine C++** and **Blueprint**, improving gameplay scalability by **30%**.
- Applied **advanced OOP design patterns** and component-based architecture to create reusable gameplay modules, reducing code redundancy by **25%** and developing integrated UI interactions that increased player testing engagement by **40%**.
- Collaborated in a **20-member** team using **Agile methodology**, managing tasks via **Git and issue tracking**, while optimizing game performance to boost frame rate stability by **~20%** in prototypes

Member, HOFSTRA C.O.D.E

September 2024 – Present

Hofstra University

- Participate in weekly coding challenges and hackathons.
- Engage in peer-led workshops to explore emerging technologies.